Journal of Materials Chemistry A

ROYAL SOCIETY OF CHEMISTRY

COMMUNICATION

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Cite this: J. Mater. Chem. A, 2020, 8, 12314

Received 25th April 2020
Accepted 1st June 2020

DOI: 10.1039/d0ta04360e

rsc.li/materials-a

Published on 01 June 2020. Downloaded on 11/17/2025 7:20:10 AM.

Zinc ion trapping in a cellulose hydrogel as a solid electrolyte for a safe and flexible supercapacitor†

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A flexible zinc-ion hybrid supercapacitor (ZHS) was constructed using a solid-state cellulose hydrogel electrolyte containing a high concentration of $ZnCl_2$. This biodegradable cellulose hydrogel electrolyte can afford a wide electrochemical potential window and excellent ionic conductivity, resulting in a ZHS with an extremely high capacity of 193 mA h g^{-1}, an energy density of 192 W h kg^{-1}, and a power density of 16 976 W kg^{-1}. The flexible ZHS also shows excellent stability and can work at low temperature of −20 °C. This work progresses the field of Zn-based electrochemical energy storage devices and gives a meaningful guide for the design of solid-state electrolytes.

Due to the rapid development of wearable electronic devices, flexible electrochemical energy storage devices (EESDs) with high energy/power densities are urgently needed.[1,2] Solid-state electrolytes are a key component for assembling a flexible EESD and afford the charge carriers. Additionally, they can replace the separator to prevent a short circuit between the two electrodes.[3] Considering the rapid advancement of the electronics industry, there is an urgent need to develop renewable electrolytes, which are far behind the development of renewable electrode.[4,5] Cellulose, the most copious natural polymer in the world, is non-toxic, physically/chemically stable, biodegradable, and has been used as an ideal source for solid/gel polymer electrolytes.[5–14] To support EESDs with high capacitance performances, gel polymer electrolytes are attractive because they can trap liquid electrolytes inside the polymer matrix and thus present much higher ionic conductivities than solid polymer electrolytes.[6,7,15] Gel polymer electrolytes, with cellulose as the host, are completely biodegradable and sustainable. However, it is difficult to have a pure cellulose matrix with both high ionic conductivity and mechanical integrity.[5,7]

Zinc-ion EESD with aqueous electrolytes is a promising candidate material for wearable devices because of its safety, environmental friendliness, and low cost.[16–18] The zinc anode has a high theoretical capacity (823 mA h g^{-1}) and low redox potential (−0.76 V relative to a standard hydrogen electrode).[19] Zinc-ion hybrid supercapacitors (ZHSs) combine the advantages of both supercapacitors and batteries and have become a high-profile research area in recent years due to their high energy and power densities.[9,20–29] Dong *et al.* reported a ZHS with a $ZnSO_4$ aqueous solution, zinc foil, and activated carbon as the electrolyte, anode, and cathode, respectively.[20] This ZHS presented a capacity of 121 mA h g^{-1} and a high-power output of 14 900 W kg^{-1}. Flexible solid-state ZHSs were also studied and the authors focused on electrolytes such as $ZnSO_4$-included aqueous gelatin or polymer hydrogels.[9,25,26,28,29] The ionic conductivity needs to be specifically considered in zinc ion-containing electrolytes.[2,16,18] Furthermore, ZHSs with water-based electrolytes would inevitably encounter water splitting reactions and constraints to the stability of the electrochemical potential window (EPW).[18,30]

For achieving electrochemical stability and a broad potential window, water-in-salt electrolytes are proposed.[31,32] A water-in-salt $ZnCl_2$ was used to improve the performance of a Zn-based rechargeable battery, where the operable EPW was widened and the cycle life was extended.[33–36] The potential for oxygen evolution reactions was raised. The water-in-salt $ZnCl_2$ and Zn^{2+} ions showed near-ideal depositing and stripping on the zinc metal anode.[35,36] $ZnCl_2$ is cheap and highly soluble, thus showing the potential in water-in-salt electrolyte to improve the performance of ZHSs. Herein, we designed an extraordinary strategy to trap a high concentration of $ZnCl_2$ in a cellulose matrix as a gel polymer electrolyte.[37] The gel polymer electrolyte (cellulose hydrogel) was obtained by dissolving cotton cellulose in a concentrated $ZnCl_2$ solution. After adding a trace amount of calcium ions, both Zn^{2+} and Ca^{2+} would be locked in the cellulose hydrogel *via* metal–cellulose interactions. A flexible ZHS was then fabricated using the

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† Electronic supplementary information (ESI) available. See DOI: 10.1039/d0ta04360e

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cellulose hydrogel as the electrolyte. It was shown to be quite stable and can endure at a low temperature of −20 °C. The ZHS achieved an excellent energy density of 192 W h kg^{-1} at a power density of 499 W kg^{-1}, and 58 W h kg^{-1} at an extremely high-power density of 16 976 W kg^{-1}.

As shown in Fig. 1, a cellulose hydrogel with a high concentration of $ZnCl_2$ (1.75 g g^{-1} $ZnCl_2/H_2O$) was designed and used as a solid-state electrolyte in zinc-ion hybrid supercapacitors. The cellulose possesses the optimal solubility of $ZnCl_2$ with a specific hydrate composition ($[Zn(OH_2)_6][ZnCl_4]$), and the adjacent Zn–cellulose junction zones were connected by Ca^{2+} ions as the ion crosslinking agent.[38] The cellulose hydrogel has good resistance to drying because of its coordination structure.[37] It is colorless and transparent, suggesting that the $ZnCl_2$ is highly dispersed in the cellulose. Furthermore, it is highly flexible and stable even at low temperatures (*e.g.* −60 °C).[37] The cellulose hydrogel can be stretched to nearly 100% under 0.37 MPa (Fig. S1, ESI†). To fabricate the flexible ZHS, the hydrogel was assembled with a zinc foil anode and an activated carbon cathode. The hydrogel electrolyte provides highly concentrated charge carriers for high-power reversible energy storage. During the charge/discharge process, the ions are adsorbed/desorbed on the carbon electrode and the main Zn^{2+} is deposited/striped on the zinc foil electrode.[20,21,33]

The electrochemical performance of the solid-state zinc-ion hybrid supercapacitor was studied. Cyclic voltammetry (CV) was performed at different voltage windows between 0 and 1.6–2.2 V at 50 mV s^{-1}. As shown in Fig. 2a, a perfectly symmetrical CV curve was obtained at a voltage of as high as 0–2.0 V, and there was no obvious current surge until the positive potential increases to 2.2 V. Moreover, the CV curves at various scan rates were also tested and they are all approximately rectangles even at a high scan rate of 200 mV s^{-1}, suggesting the highly reversible nature of the capacitor (Fig. 2b). Fig. 2c shows the galvanostatic charge discharge (GCD) curves of ZHS at different current densities. According to the GCD results, the discharge capacity could reach an extremely high value of 193 mA h g^{-1} at 0.5 A g^{-1} and remained at 68 mA h g^{-1} at a high current density of 20 A g^{-1}. At a current density of 0.5–20 A g^{-1}, the coulombic efficiencies exceed 90.0% with a maximum of 99.4% at 5 A g^{-1}. The energy/power densities were calculated (Fig. 2d). Our ZHS could achieve an energy density of

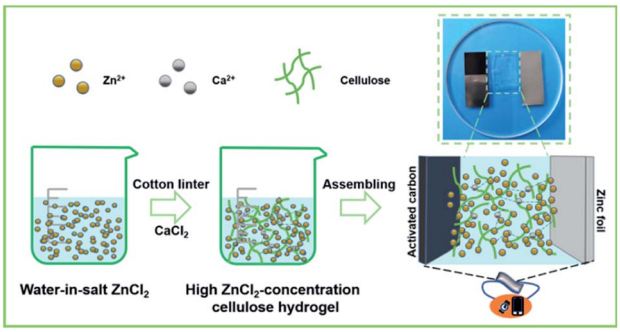

Fig. 1 Schematic of the fabrication of the cellulose hydrogel containing a high concentration of $ZnCl_2$ and its application as a zinc-ion hybrid supercapacitor.

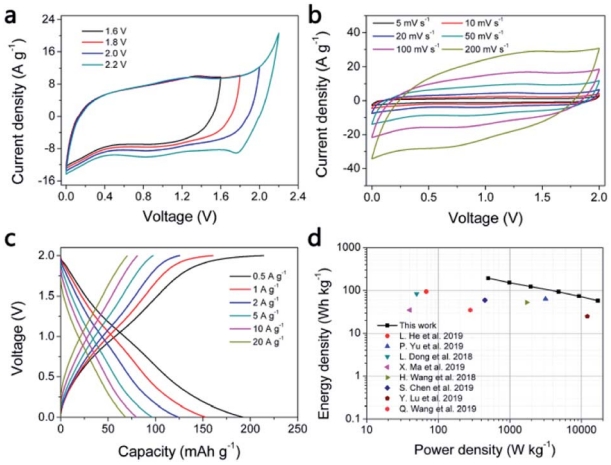

Fig. 2 CV curves in different voltage windows at 50 mV s^{-1} (a) and at different scan rates from 5 to 200 mV s^{-1} (b), GCD curves at different current densities from 0.5 to 20 A g^{-1} (c), and a Ragone plot of the flexible ZHS compared with other zinc-ion hybrid supercapacitors (d).

192 W h kg^{-1} at a power density of 499 W kg^{-1} and 58 W h kg^{-1} at 16 976 W kg^{-1}. In literature, Chen *et al.* reported that a flexible ZHS with a $ZnSO_4$-containing polyacrylamide hydrogel electrolyte could reach an energy density of 59.7 W h kg^{-1} at 447.8 W kg^{-1}.[25] The flexible ZHS assembled with a gelatin/$ZnSO_4$ gel electrolyte showed a maximum energy density of 86.8 W h kg^{-1} at 429.6 W kg^{-1}.[26] Obviously, great progress has been made in our ZHS, where the performance is better than that of the other zinc-ion hybrid supercapacitors.[20,21,24–26,39–41]

The superior capacitive performance of our ZHS can be attributed to the use of the cellulose hydrogel with a high concentration of $ZnCl_2$. The presence of $CaCl_2$ mainly enhances the strength of the hydrogel. The water molecules in the hydrogel are restrained due to their limited content and the hydrated state of $ZnCl_2$. Therefore, an increase in the onset potential of oxygen evolution is possible when Zn^{2+} depositing occurred to obtain a wider electrochemical potential window (2.0 V). The good ionic conductivity of the cellulose hydrogel also needs to be mentioned. In general, a salt solution with an extremely high concentration is highly viscous and shows poor ion mobility. The water-in-salt $ZnCl_2$ shows decreased ionic conductivity with increasing ion-concentration.[33,42] However, the ionic conductivity of the cellulose hydrogel with a high concentration of $ZnCl_2$ reached a high value of 74.9 mS cm^{-1} (Fig. S2, ESI†).[37] In the cellulose hydrogel, the cellulose is highly dispersed and thus would act as a dispersion medium for the ions, allowing the ions to maintain a good migration ability at high concentrations. The ionic conductivity of the cellulose hydrogel is much higher than that of the water-in-salt $ZnCl_2$ (22.8 mS cm^{-1}) solution at the same concentration (1.75 g g^{-1} $ZnCl_2/H_2O$). The ionic conductivity of our cellulose hydrogel was also compared to some others,[5,9,15,26,33,42] including various gel electrolytes and aqueous $ZnCl_2$ and Zn_2SO_4. Among the above examples, our electrolyte shows the highest ionic conductivity (Table S1, ESI†). For comparison, ZHS with 1.75 g g^{-1} $ZnCl_2/H_2O$ water-in-salt $ZnCl_2$ as the liquid electrolyte was

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assembled. The nonwoven cloth was used to retain the $ZnCl_2$ electrolyte and as a separator between the zinc foil anode and activated carbon cathode. At a low current density of 0.5 A g^{-1}, the ZHS held a comparable capacity of 188 mA h g^{-1}, whereas at a high current density of 20 A g^{-1}, the capacity drastically reduced to 34 mA h g^{-1}, which is only half of that of the cellulose hydrogel electrolyte-based ZHS (CV and GCD curves in Fig. S3 and S4, ESI†). The ionic conductivity is important for ion storage and contributes to high capacitive performance.

The cycling performance of the ZHS was tested at a current density of 5 A g^{-1}. As shown in Fig. 3a, the capacity retention is 94.7% after 5000 charge/discharge cycles and the coulombic efficiency is maintained at almost 100%. Fig. 3b compares the XRD patterns of the zinc foil anode before and after 5000 cycle tests, and only pure zinc metal can be seen. Fig. 3c and d show the SEM images of the corresponding zinc foil. After the cycles, the surface of the zinc foil anode became rough and no dendrite could be found. These results suggest the high reversibility of Zn^{2+} during the depositing/stripping process in the hybrid capacitor energy storage system. The extremely low water content is tightly bonded with Zn^{2+} and thus reactions with the Zn electrode to form irreversible products, such as $Zn(OH)_2$ and ZnO, which are bad for cycle performance, are restricted.[33,35] A high concentration of Zn^{2+} in the cellulose hydrogel contributes to a low gradient for the uniform and stable deposition/stripping of Zn.[36] The speciation of ions of the high $ZnCl_2$-concentrated cellulose hydrogel was studied using Raman spectroscopy and comparing with a 1.75 g g^{-1} $ZnCl_2/H_2O$ water-in-salt $ZnCl_2$ solution.[33] Both the hydrogel and water-in-salt $ZnCl_2$ solution show three peaks at around 108, 284, and 382 cm^{-1}. The two peaks at around 108 and 284 cm^{-1} can be assigned to $[Zn(OH_2)_2Cl_4]^{2-}$. The broad peak at around 382 cm^{-1} is much weaker than the peaks for $[Zn(OH_2)_2Cl_4]^{2-}$, suggesting the minor presence of $[Zn(OH_2)_6]^{2+}$ (Fig. S5, ESI†). The anions participate in reversible adsorption/desorption in the form of $[Zn(OH_2)_2Cl_4]^{2-}$. Meanwhile, Zn^{2+} deposition/stripping on the anode is accompanied by a small number of $[Zn(OH_2)_6]^{2+}$.[20]

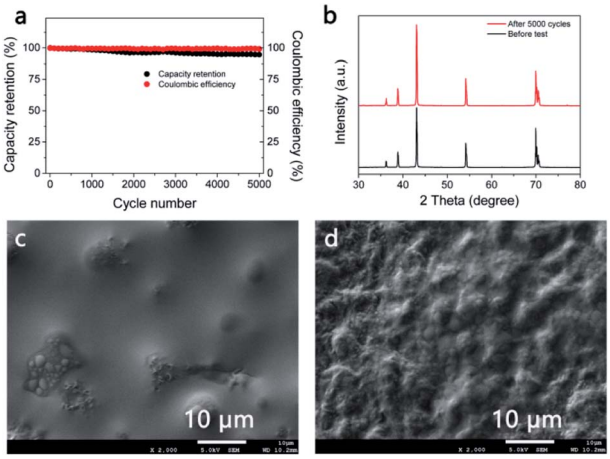

Fig. 3 Cycling stability at a current density of 5 A g^{-1} (a), XRD patterns (b) and SEM image of the zinc foil anode before (c) and after (d) 5000 cycle tests.

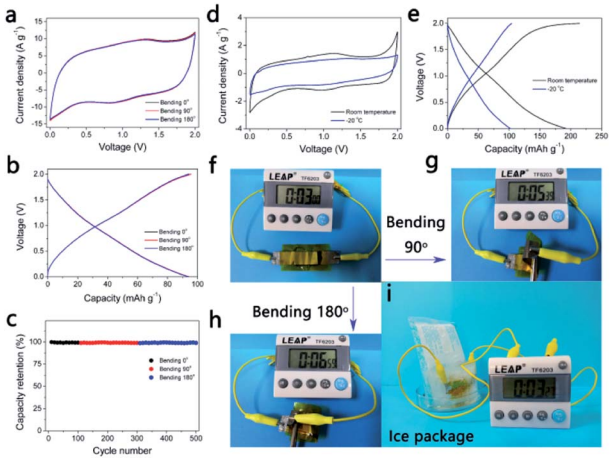

Fig. 4 The CV curves (a) at different bending angles with a scan rate of 50 mV s^{-1}; GCD curves (b) at different bending angles with a current density of 5 A g^{-1}; cycling stability at 5 A g^{-1} under various bending conditions (c); CV curves (d) at a low temperature of −20 °C at 5 mV s^{-1}; GCD curves (e) at −20 °C at 0.5 A g^{-1}; The flexible zinc-ion hybrid supercapacitor powers a timer at different bending angles of 0° (f), 90° (g), 180° (h), and at −20 °C (i).

The flexibility of the cellulose hydrogel-based ZHS was examined. Fig. 4a and b show that the CV and GCD curves of the ZHS can be reproduced with the same capacitive behavior at different bending angles. Fig. 4c presents the ability of the system work continuously under bending conditions. After 500 cycle tests with various degrees of bending, the ZHS maintains almost the same capacity as its initial one. Considering that the concentrated inorganic salts can inhibit water from freezing, the capacitive behavior of this ZHS using cellulose hydrogel containing a highly concentrated $ZnCl_2$ were also studied at a low temperature of −20 °C.[28,43] Compared to the room-temperature test, the area of the CV curve was slightly reduced when testing was performed at −20 °C (Fig. 4d). The GCD curve also indicates a reduced performance but the capacity remains quite high with a value of 102 mA h g^{-1} at the current density of 0.5 A g^{-1} (Fig. 4e). The CV curve maintained an approximately rectangular shape at a scan rate of 200 mV s^{-1} and the GCD curve shows a capacity of 22 mA h g^{-1} at a high current density of 20 A g^{-1} (Fig. S6 and S7, ESI†). Therefore, high-power behavior under −20 °C is achievable with this ZHS. When comparing the Nyquist plot between −20 °C and room temperature, the increase in charge transfer resistance is more conspicuous than the serial internal resistance, indicating that the low-temperature capacitive performance was mainly due to the ionic conductivity of cellulose hydrogel (Fig. S8, ESI†). The cellulose hydrogel can retain an ionic conductivity of 47.7 mS cm^{-1} at −20 °C (Fig. S2, ESI†). Fig. 4f–i demonstrate the practical application of our device to power a timer. Under different bending conditions, the ZHS can supply a timer (more information in Fig. S9, ESI†). After ice package at a low temperature of −20 °C, the ZHS still worked, revealing the actual low temperature resistance (Fig. 4i). It is worth mentioning that the zinc-ion hybrid supercapacitor with

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a cellulose hydrogel electrolyte can also endure high temperatures and even presents enhanced capacity at 50 °C (Fig. S10, ESI†).

Conclusions

In conclusion, a novel strategy was designed to trap a high concentration of $ZnCl_2$ in a cellulose hydrogel. This cellulose hydrogel was used as a solid-state electrolyte in flexible ZHS, achieving extraordinary capacitive performances. A stable EPW of 2.0 V and excellent ionic conductivity contribute to a ZHS with a capacity of 193 mA h g^{-1} at a current density of 0.5 A g^{-1} and an energy density of 192 W h kg^{-1} at a power density of 499 W kg^{-1}. This ZHS shows a stable cycle performance with 94.7% capacity retention after 5000 cycles and a low temperature tolerance (−20 °C). This study opens up a door for the use of sustainable cellulose hydrogel electrolytes in the field of electrochemical energy storage.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

The authors are grateful for the financial support of the Youth Fund of Natural Science Foundation of Jiangsu Province (BK20170919, BK20180766) and the National Natural Science Foundation of China (21808112).

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